

electromagnetic fields 14, 121
 electromagnetic force 20, 22, 119, 123, 126, 134, 139
 electromagnetic radiation 14, 16, 26–27
 electromagnetic repulsion 20, 22, 69
 electromagnetic spectrum 14
 electromagnetic waves 25
 electron detectors 100, 101
 electron microscopes 100–101
 electron orbitals 31
 electronics 85, 90–91
 electrons 12, 22, 46–47, 112
 accelerated 100
 antimatter 128–29
 atomic clocks 88
 atomic structure 10–11
 band theory of solids 90
 clouds 11
 Cooper pairs 81, 94, 95
 energy states 30, 31, 69, 83, 94
 Higgs field 127
 jumping between energy levels 30, 88
 LEDs 92
 magnetic moments 67
 photoelectric effect 32–33
 quantum electrodynamics 134
 quantum field theory 132
 Rydberg 83
 size 8
 solid-state physics 76–77
 spin state 38, 106
 stimulated emission 88
 superconductivity 81
 superposition 38–39
 transistors 91
 wavelike properties 18–19
 wavelength 100
 electroweak force/theory 139

elementary particles 12, 68, 119, 122, 123, 124–25
 elements
 atoms 10
 creating 116–17
 energy
 antimatter 128–29
 nuclear fission 114–15
 nuclear fusion 116–17
 photosynthesis 148–49
 energy quantization 28–29
 energy shells 10, 30, 31, 69
 energy states 30, 78, 86, 88, 90, 94, 148
 entanglement *see* quantum entanglement
 environmental factors 60, 75
 enzymes 151
 equilibrium point 36
 Everett, Hugh III 54
 excited states 30, 86, 87

F

feedback electronics 102
 fermions 12, 68, 69, 76, 81, 119, 124–25, 130, 134
 Feynman diagrams 134
 fission, nuclear 114–15
 force carrying particles 119, 124–25, 126
 Fourier transforms 41
 frequency
 microwave radiation 88, 89
 wave function 41
 fundamental forces 20–23, 125, 126, 137, 138–9, 142, 144
 fusion, nuclear 116–17

G

gamma decay 113
 gamma rays 113
 gauge bosons 68, 124, 125, 126

General Relativity 23, 133, 137, 138, 140
 genetic mutations 147, 150
 gluons 13, 20, 122, 125, 126, 135
 grand unified theories 138–39
 gravitational fields 144, 145
 gravitons 23, 142
 gravity 23, 130, 131, 140–41
 theories of everything 138–39, 142–45
 green light 33
 ground states 30, 86

H

hadrons 12, 13, 122
 half-life 113
 Hamiltonian operator 44
 Heisenberg, Werner 42–43
 helium 116
 hidden variables 56–57, 60
 Higgs bosons 13, 66, 124–25, 127, 133
 Higgs field 127, 133
 hydrogen 83, 99, 116, 128–29

IJK

infrared radiation 14, 27
 insulation 76
 interference 16, 18–19, 58
 intrinsic angular momentum 66, 67
 ions 76, 81, 83, 87, 109
 Josephson junctions 95, 96, 97
 kinetic energy 82

L

Large Hadron Collider 121
 lasers 82, 87, 102, 103, 109
 lattice, geometric 76–77, 81, 94, 109